Naive Bayes

Are they flu or not? Use Naive Bayes. Chills=Y, Runny Nose=N, Headache=Mild, Fever=Yes, Flu=?

Chills	Runny Nose	Headache	Fever	Flu
Y	N	Mild	Y	0
Y	Y	No	N	1
Y	N	Strong	Y	1
N	Y	Mild	Y	1
N	N	No	N	0
N	Y	Strong	Y	1
N	Y	Strong	N	0
Y	Y	Mild	Y	1

PCA

$[1\ 1]^T$ $[2\ 2]^T$ $[3\ 3]^T$

Use PCA to map given 2D points to 1D space.

1. What is the first principal compenent?
2. Use your PCA to map data points to 1D space. What is the variance of them?
3. What will be the reconstruction error if you want to go back to 2D from 1D

SVM

A kernel functions is given. Show that it maps R^2 to R^5 (R^5 matrix is given)

$$K(x, y) = (x.y + 1)^2 + 2(x, y)$$

I don't remember the exact kernel function but it was looking like that.

K-Means

Part A

Explain sum-of-squares for k-means algorithm. How the error is minimized iteratevily. Explain k-means clustering algorithm or give pseudocode.

Part B

A plot of sum-of-squares and number of clusters is given. How many clusters does the dataset have? Why?

Multiple Choice Part

Overfitting, regularization, how to avoid overfitting

Sınav 8 sayfaydı. 5 soru vardı. İstediğin 4 tanesini seçiyorsun diğeri bonus oluyor. Çoktan seçmeli kısım vardı bir sorunun birden fazla şıkkının doğru olabildiği. Cheatsheet yok. Hesap makinesi var.